

Relational Model, ER Diagrams & SQL - Detailed Lecture Notes

DETAILED COMPREHENSIVE NOTES

Subject: Database Management Systems | Chapter: Chapter 1: Relational Model, ER Diagrams & SQL Fundamentals | Academic Year: 2023-2024

1. Database Architecture & Three-Schema Architecture

A Database Management System (DBMS) is software designed to store, manage, and retrieve structured data securely and concurrently.

The ANSI-SPARC Three-Schema Architecture provides physical data independence (ability to alter physical storage without changing conceptual schema) and logical data independence (ability to alter conceptual schema without altering user views).

Internal Level: Physical storage structures, indexes, block allocations.

Conceptual Level: Entities, data types, relationships, constraints.

External Level: Individual end-user views and application interfaces.

2. Entity-Relationship (ER) Modeling

ER modeling visualizes data structures. Entities are real-world objects represented by rectangles. Attributes are properties represented by ellipses (Key attributes underlined, Multivalued in double ellipses, Derived in dashed ellipses).

Relationships connect entities (represented by diamonds) with cardinality constraints: One-to-One (1:1), One-to-Many (1:N), and Many-to-Many (M:N).

- * Strong Entity: Has a primary key and independent existence.
- * Weak Entity: Cannot be uniquely identified by its own attributes; depends on a strong identifying entity via foreign key and partial discriminator key (dashed underline).

3. SQL Data Definition & Query Language

DDL (Data Definition Language): CREATE, ALTER, DROP, TRUNCATE. These commands define and modify database schema structures and are auto-committed.

DML (Data Manipulation Language): INSERT, UPDATE, DELETE. These manipulate data records and require COMMIT/ROLLBACK.

DQL (Data Query Language): SELECT with WHERE, GROUP BY, HAVING, ORDER BY clauses. The logical processing order is FROM -> WHERE -> GROUP BY -> HAVING -> SELECT -> ORDER BY -> LIMIT.

Joins: INNER JOIN returns matching rows; LEFT/RIGHT OUTER JOIN retains unmatched rows from left/right table; FULL OUTER JOIN retains all rows.